



Faculty of Computer Science and Information Technology

***ECHOZY: WEB-BASED MULTILINGUAL AUGMENTATIVE AND
ALTERNATIVE COMMUNICATION (AAC) PLATFORM FOR STROKE
PATIENTS WITH COMMUNICATION IMPAIRMENTS***

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**Bachelor of Computer Science with Honours
(Software Engineering)
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INSTALLATION KIT & USER MANUAL INFORMATION

Item	Details
Prepared by	Fatin Farhanah Binti Halidin
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ABSTRACT

Stroke patients often experience communication impairments that significantly affect their daily interactions and rehabilitation process. Existing Augmentative and Alternative Communication (AAC) solutions are frequently limited by platform dependency, lack of localization, high cost, and limited adaptability to users' communication habits. This project proposes Echozy, a web-based multilingual AAC platform designed to support Malaysian stroke patients through touch-based interaction, Bahasa Malaysia and English communication content, online text-to-speech communication, and personalization of frequently used phrases and vocabulary. The system also enables caregivers and healthcare providers to manage patient profiles, customize patient-specific phrases and vocabulary, monitor communication usage, view practice progress, and track patient session activity through role-based access. In addition, admins are able to manage default phrases and vocabulary to support consistent global content across the system. Echozy was developed using the Rapid Application Development (RAD) methodology, which involved requirement gathering, prototype evaluation, iterative design, and system refinement. The system design is supported by use case diagrams, sequence diagrams, a class diagram, an entity-relationship diagram, and a normalized database design up to Third Normal Form (3NF). The proposed system aims to provide an accessible, user-centred, and locally relevant communication solution that supports effective interaction, patient-specific communication, and rehabilitation support for stroke patients, particularly within the Malaysian context.

Keywords: Augmentative and Alternative Communication (AAC), stroke patients, communication impairments, multilingual communication, text-to-speech, usage-based personalization, web-based system

ABSTRAK

Pesakit strok sering mengalami gangguan komunikasi yang memberi kesan besar terhadap interaksi harian dan proses pemulihan mereka. Kebanyakan sistem Augmentative and Alternative Communication (AAC) sedia ada mempunyai kekangan dari segi kebergantungan kepada platform tertentu, kekurangan sokongan bahasa tempatan, kos yang tinggi, serta keterbatasan dalam menyesuaikan diri dengan tabiat komunikasi pengguna. Projek ini mencadangkan Echozy, sebuah platform AAC berasaskan web dan pelbagai bahasa yang direka untuk menyokong pesakit strok di Malaysia melalui interaksi sentuhan, kandungan komunikasi dalam Bahasa Malaysia dan Bahasa Inggeris, komunikasi teks-ke-ucapan secara dalam talian, serta pemeribadian frasa dan kosa kata yang kerap digunakan. Sistem ini juga membolehkan penjaga dan penyedia perkhidmatan kesihatan mengurus profil pesakit, menyesuaikan frasa dan kosa kata khusus pesakit, memantau penggunaan komunikasi, melihat kemajuan latihan, serta menjejaki aktiviti sesi pesakit melalui akses berasaskan peranan. Selain itu, pentadbir boleh mengurus frasa dan kosa kata lalai bagi menyokong kandungan global yang konsisten dalam sistem. Echozy dibangunkan menggunakan metodologi Rapid Application Development (RAD), yang melibatkan pengumpulan keperluan, penilaian prototaip, reka bentuk secara berulang, dan penambahbaikan sistem. Reka bentuk sistem disokong oleh rajah kes guna, rajah jujukan, rajah kelas, rajah hubungan entiti, serta reka bentuk pangkalan data yang dinormalisasikan sehingga Third Normal Form (3NF). Sistem yang dicadangkan ini bertujuan untuk menyediakan penyelesaian komunikasi yang mudah diakses, berpusatkan pengguna, dan relevan dengan konteks tempatan bagi menyokong interaksi berkesan, komunikasi khusus pesakit, serta sokongan pemulihan untuk pesakit strok, khususnya dalam konteks Malaysia.

Kata kunci: *Augmentative and Alternative Communication (AAC), pesakit strok, gangguan komunikasi, komunikasi pelbagai bahasa, teks-ke-ucapan, pemeribadian berasaskan penggunaan, sistem berasaskan web*

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LIST OF ABBREVIATIONS

AAC	Augmentative and Alternative Communication
FCSIT	Faculty of Computer Science and Information Technology
FYP	Final Year Project
RAD	Rapid Application Development
TTS	Text-to-Speech
UNIMAS	Universiti Malaysia Sarawak

CHAPTER 1

INTRODUCTION

Echozy is a web-based multilingual **Augmentative and Alternative Communication (AAC)** platform developed to support stroke patients with communication impairments. The system allows patients to communicate using touch-based phrase and vocabulary cards, Bahasa Malaysia and English content, message building, and online text-to-speech output.

This document combines the Installation Kit and User Manual for the Echozy system. The Installation Kit explains how to prepare, configure, run, and publish the system. The User Manual explains how caregivers, healthcare providers, admins, and patients can use the system according to their roles.

Echozy is implemented using HTML, CSS, and JavaScript for the frontend. The backend services are supported by Supabase Authentication, Supabase PostgreSQL Database, Supabase Storage, and Supabase Edge Functions. OpenAI Text-to-Speech API is used for online speech generation, GitHub is used for version control, and Vercel is used for hosting and deployment.

CHAPTER 2

INSTALLATION KIT

2.1 Installation Kit Overview

The Echozy installation kit provides the required project files, configuration steps, external service links, and deployment instructions needed to run or publish the system. Since Echozy is a web-based system, it does not require a mobile app installer. Users can access the deployed system through a browser, while developers can run and maintain the source code through GitHub, Supabase, OpenAI, and Vercel.

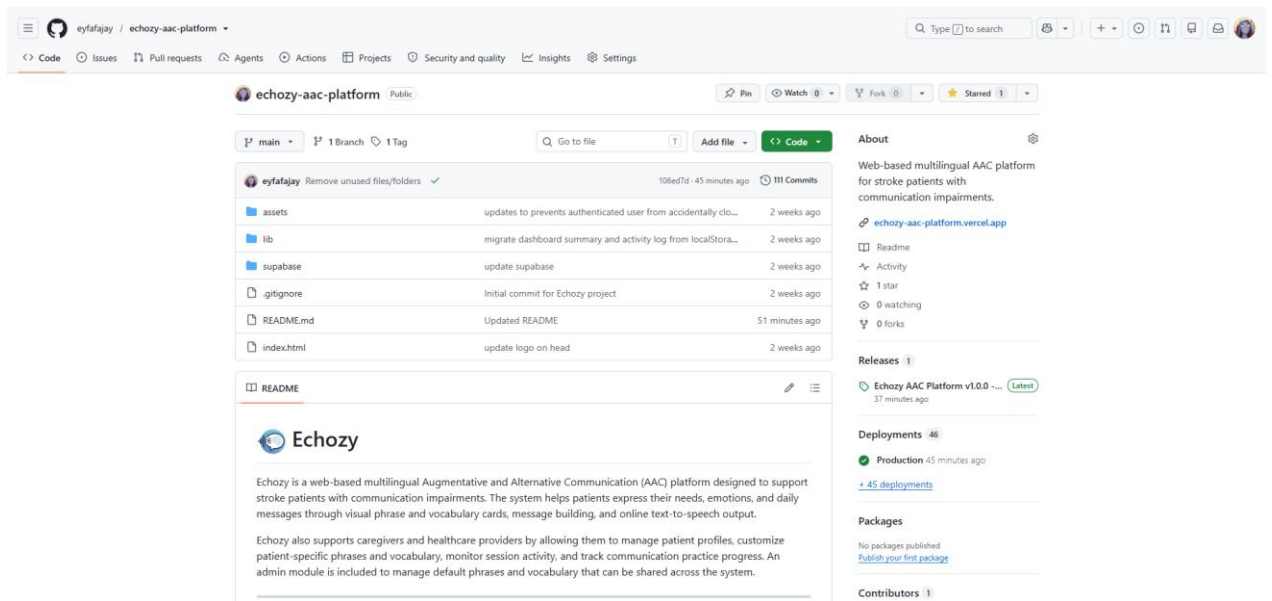


Figure 1K-1: Echozy GitHub Repository Page for Echozy System

Figure IK-1 shows the Echozy GitHub repository, which stores the source code, project files, and version control history for the system.

2.2 Project Links and External Service Links

Table IK-1 lists the main project and service links required to access, run, configure, or publish the Echozy system.

Table IK-1: Main Project and Service Links

Link Name	URL	Purpose
GitHub Repository	https://github.com/eyfafajay/echozy-aac-platform	Access and download the Echozy source code.
Hosted Echozy Website	https://echozy-aac-platform.vercel.app	Access the deployed Echozy web application.
Supabase Dashboard	https://supabase.com/dashboard	Manage Supabase project, database, authentication, storage, and edge functions.
Supabase Documentation	https://supabase.com/docs	Reference for Supabase services and setup.
OpenAI Text-to-Speech Documentation	https://platform.openai.com/docs/guides/text-to-speech	Reference for OpenAI speech generation integration.
OpenAI API Keys	https://platform.openai.com/api-keys	Create or manage OpenAI API keys for TTS integration.
OpenAI API Pricing	https://openai.com/api/pricing/	Monitor possible long-term TTS cost planning.
Vercel Dashboard	https://vercel.com/dashboard	Manage hosting and deployment.
Vercel Git Deployment Documentation	https://vercel.com/docs/git	Reference for GitHub and Vercel deployment workflow.
GitHub Release Documentation	https://docs.github.com/en/repositories/releasing-projects-on-github/about-releases	Reference for publishing project release versions.

2.3 System Requirements

Table IK-2 shows the recommended requirements to run and maintain the Echozy system.

Table IK-2: Recommended Requirements for Echozy System

Requirement Type	Recommended Requirement
Operating System	Windows, macOS, or Linux
Code Editor	Visual Studio Code
Browser	Google Chrome, Microsoft Edge, Safari, or any modern web browser
Internet Connection	Required for Supabase authentication, database access, storage access, Vercel hosting, and online TTS.
Git	Required to clone, commit, push, and manage the GitHub repository.
Supabase Account	Required for Authentication, PostgreSQL Database, Storage, and Edge Functions.
OpenAI API Account	Required to generate an API key for online text-to-speech output.
Vercel Account	Required for hosting and deployment.

2.4 Installation Package Contents

Table IK-3 describes the expected installation package contents for Echozy. The exact folder names may vary depending on the final repository structure, but the package should include the frontend files, Supabase configuration, assets, and documentation.

Table IK-3: Expected Installation Package Contents for Echozy

Item	Description
README.md	Project overview, setup notes, technology stack, and usage information.
index.html	Main homepage file for the Echozy web application.
assets/	Stores CSS, JavaScript, images, icons, logos, and other frontend assets.
lib/	Stores role-based pages, authentication pages, dashboard pages, patient pages, and admin pages.
supabase/functions/tts/	Stores the Supabase Edge Function used for OpenAI Text-to-Speech integration.
.gitignore	Prevents unnecessary or sensitive files from being committed to GitHub.
Documentation files	Installation kit, user manual, thesis-related documentation, screenshots, and report files.

```
echozy-aac-platform/  
├── README.md  
├── index.html  
├── assets/  
│   ├── css/  
│   ├── js/  
│   └── images/  
├── lib/  
│   ├── auth/  
│   ├── main/  
│   ├── patients/  
│   └── admin/  
├── supabase/  
│   └── functions/  
│       └── tts/  
└── .gitignore
```

2.5 Local Installation Steps

This section explains the steps to download and run Echozy locally for development or review.

1. Open the GitHub repository link: <https://github.com/eyfafajay/echozy-aac-platform>
2. Click Code, then copy the HTTPS repository link.
3. Open Visual Studio Code and open a terminal.
4. Clone the repository using the Git command shown below.

```
git clone https://github.com/eyfafajay/echozy-aac-platform.git
```

5. Open the project folder.

```
cd echozy-aac-platform
```

6. Open the folder in Visual Studio Code.

```
code .
```

7. Check the Supabase URL, Supabase anon key, Supabase Storage paths, and Edge Function URL used in the JavaScript files.
8. Run the project using Live Server or any local static server.
9. Right click `index.html` → Open with Live Server

If the system opens successfully in the browser, continue with Supabase and OpenAI configuration to ensure login, database retrieval, image display, and text-to-speech functions work correctly.

2.6 Supabase Configuration

Echozy uses Supabase Authentication, PostgreSQL Database, Storage, and Edge Functions.

Figure IK-2 should show the Supabase project dashboard used for Echozy configuration.

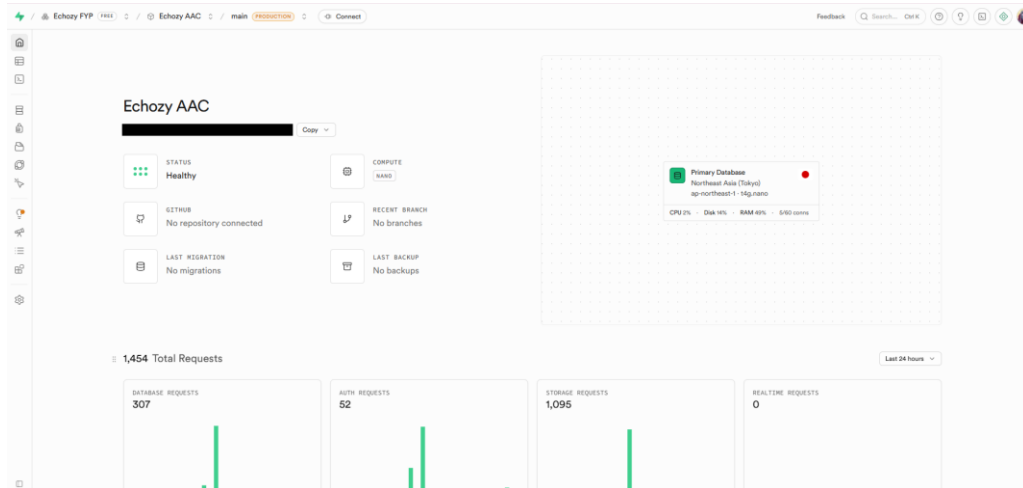


Figure 1K-2: Supabase Project Dashboard

Figure IK-2 shows the Supabase project dashboard where Authentication, Database, Storage, and Edge Functions can be configured.

2.6.1 Supabase Authentication

Supabase Authentication manages sign-up, sign-in, logout, and authenticated sessions for caregivers, healthcare providers, and admins. To configure authentication, open the Supabase dashboard, select the project, and configure Email Authentication according to the system requirements.

- Caregivers and healthcare providers can sign up and sign in.
- Admins can sign in through the admin access page.
- The profiles table stores role-related profile information and supports role-based redirection.

2.6.2 Supabase PostgreSQL Database

Supabase PostgreSQL Database stores the structured data used by Echozy. Table IK-4 lists the main database tables required by the system.

Table IK-4: Lists of Main Database Required in Echozy System

Table Name	Purpose
profiles	Stores caregiver, healthcare provider, and admin profile information.
patients	Stores patient profile details.
user_patients	Links caregivers or healthcare providers with patients.
admin_phrases	Stores default global phrase content.
admin_vocabulary	Stores default global vocabulary content.
patient_phrases	Stores patient-specific phrase content.
patient_vocabulary	Stores patient-specific vocabulary content.
usage_counts	Stores selected phrase and vocabulary click count records.
practice_results	Stores patient practice communication results.
session_logs	Stores patient session enter and quit activity.

2.6.3 Supabase Storage

Supabase Storage is used to store phrase and vocabulary images. The image file is stored in Supabase Storage, while the image path is saved in the related phrase or vocabulary database table.

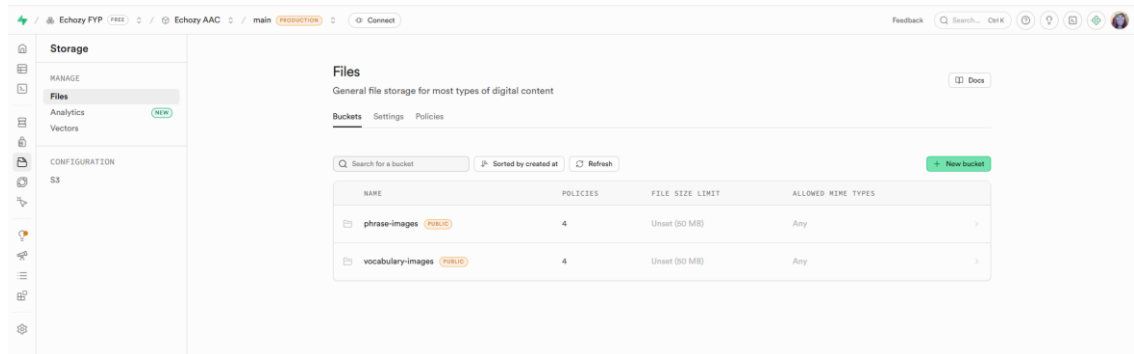


Figure 1K-3: Supabase Storage Bucket for Phrase and Vocabulary Image

Figure IK-3 shows the Supabase Storage bucket used to store phrase and vocabulary images for Echozy communication cards.

- Store admin default phrase images in the configured phrase image folder or bucket.

2.6.4 Supabase Edge Functions

Supabase Edge Functions are used as the secure backend layer for text-to-speech integration. The frontend sends selected message text to the Edge Function, and the Edge Function communicates with the OpenAI Text-to-Speech API.

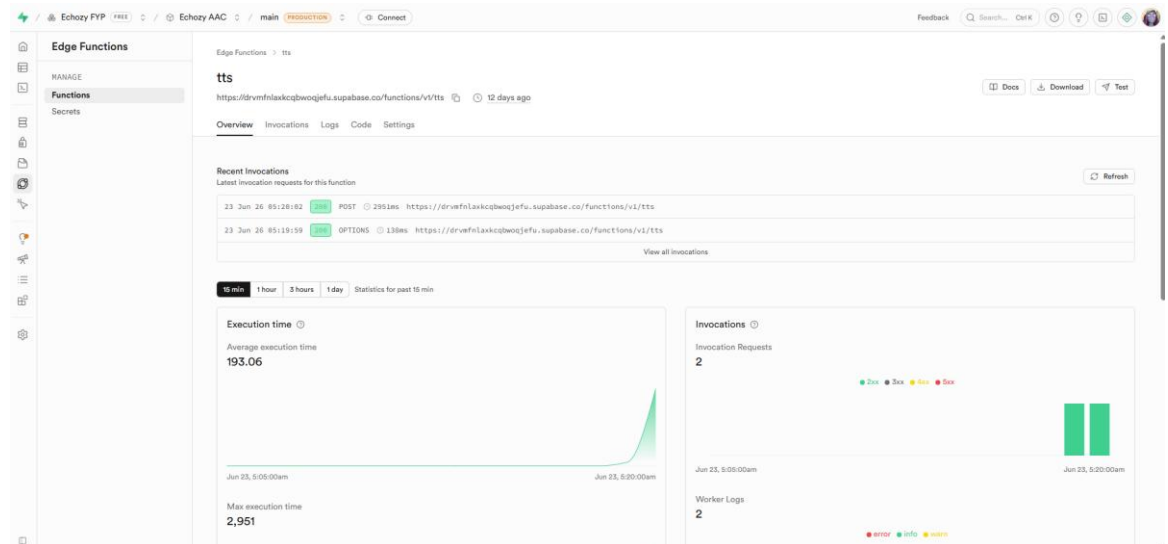


Figure 1K-4: Supabase Storage Edge Function for TTS

Figure IK-4 shows the Supabase Edge Function used for the Echozy text-to-speech integration.

```
supabase secrets set OPENAI_API_KEY=your_openai_api_key_here
supabase functions deploy tts
```

The OpenAI API key must be stored as a Supabase secret and must not be written directly inside frontend JavaScript files.

2.7 OpenAI Text-to-Speech Configuration

The OpenAI Text-to-Speech API is used to generate online speech output from selected phrase or vocabulary text. To configure the TTS feature, create an OpenAI API key, save it as a Supabase Edge Function secret, and deploy the TTS Edge Function.

1. Open the OpenAI API Keys page.
2. Create a new API key for the project.
3. Copy the API key and save it securely.
4. Store the API key in Supabase secrets using the `OPENAI_API_KEY` variable.
5. Deploy the Supabase Edge Function for TTS.
6. Test the Speak button on the Echozy Board page.

```
OPENAI_API_KEY=your_openai_api_key_here
```

For long-term use, OpenAI API usage should be monitored because online text-to-speech may involve continuous API cost when used by many patients over time.

2.8 Vercel Deployment

Vercel is used to host and deploy the Echozy web application. Figure IK-5 should show the Vercel project deployment page.

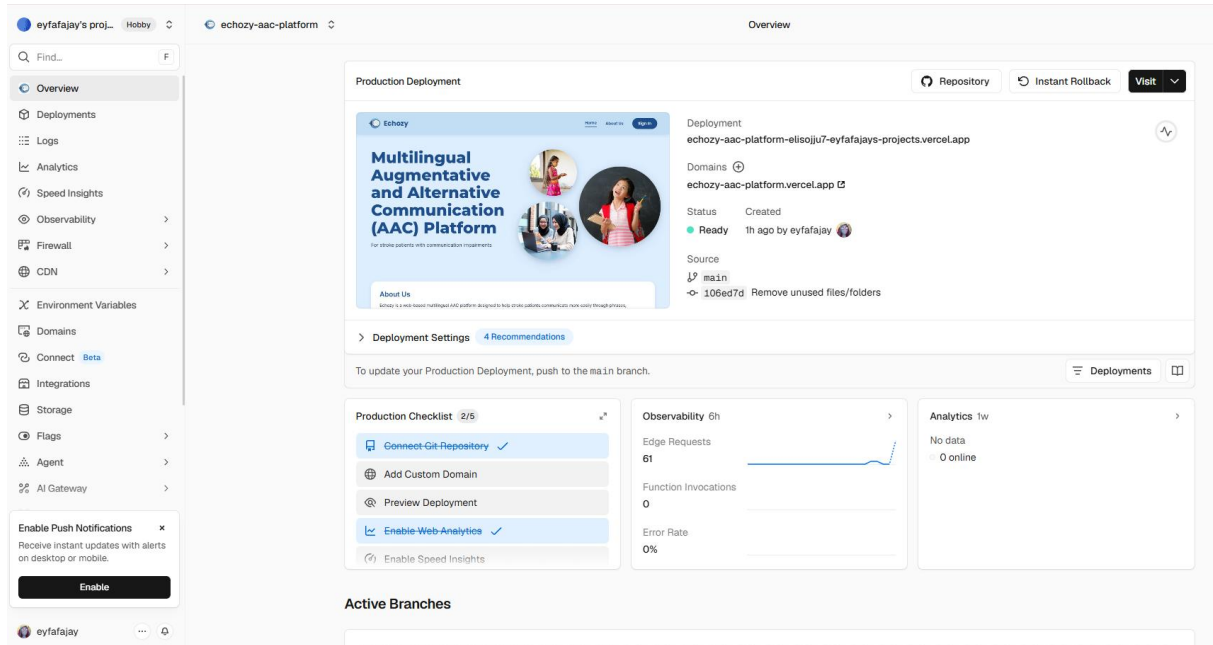


Figure 1K-5: Vercel Deployment Page for Echozy

Figure IK-5 shows the Vercel deployment page used to publish the Echozy web application.

1. Open the Vercel dashboard.
2. Import the GitHub repository: <https://github.com/eyfajay/echozy-aac-platform>
3. Select the main branch as the production branch.
4. Check the build and output settings. For a static HTML, CSS, and JavaScript project, the default static deployment setting can be used if no build command is required.
5. Add any required environment variables if the deployment uses them.
6. Click Deploy.

7. After deployment is complete, open the hosted website link: <https://echozy-aac-platform.vercel.app>
8. When changes are pushed to the connected GitHub repository, Vercel can automatically create a new deployment based on the latest code.

When changes are pushed to the connected GitHub repository, Vercel can automatically create a new deployment based on the latest code.

2.9 Installation Verification Checklist

Table IK-5 can be used to verify that the installation and deployment are working correctly.

Table IK-5: Verify Installation and Deployment Checklist

No.	Verification Item	Expected Result
1	Open hosted website	Echozy homepage loads successfully.
2	Open role selection page	Caregiver, Healthcare Provider, and Admin options are displayed.
3	Sign up as caregiver or healthcare provider	User account and profile are created successfully.
4	Sign in using valid credentials	User is redirected to the correct dashboard.
5	Add patient profile	Patient record appears in the Manage Patients page.
6	Upload phrase or vocabulary image	Image is stored and displayed correctly.
7	Enter patient session	Session log records the enter time.
8	Select phrase or vocabulary card	Selected content appears in the message builder.
9	Click Speak	Audio output is generated and played in the browser.
10	Quit patient session	Session log is updated with quit time.

2.10 Troubleshooting

Table IK-6: Troubleshooting for Echozy System

Problem	Possible Cause	Suggested Solution
Website does not load	Vercel deployment failed or URL is incorrect.	Check Vercel deployment logs and confirm the hosted URL.
Login does not work	Supabase Authentication or profile role data is not configured correctly.	Check Supabase Auth users and profiles table.
Dashboard does not redirect correctly	User role is missing or incorrect in profiles table.	Check role value for caregiver, healthcare provider, or admin.
Patient data does not appear	Database query, table relationship, or user-patient link is missing.	Check patients and user_patients tables.
Images do not display	Supabase Storage path or bucket policy is incorrect.	Check Storage bucket, file path, and image_path in database.
Speak button does not generate audio	OpenAI API key, Supabase secret, Edge Function, or request URL is not configured correctly.	Check Supabase Edge Function logs and OPENAI_API_KEY secret.
Usage count does not update	usage_counts table or update logic has an issue.	Check content ID, patient ID, content type, and update query.
Practice result does not save	practice_results table or insert logic has an issue.	Check practice result fields and database permissions.

CHAPTER 3

USER MANUAL

3.1 User Roles

Echozy supports four main user roles: stroke patient, caregiver, healthcare provider, and admin. Table UM-1 summarizes the main permissions of each role.

Table UM-1: Main Permissions of Each Role in Echozy System

User Role	Main Functions
Stroke Patient	Access communication board, select phrases and vocabulary, build messages, generate speech output, view frequently used content, and access practice communication.
Caregiver	Sign up, sign in, manage patients, manage patient phrases and vocabulary, enter and quit sessions, view activity logs, monitor usage, and update settings.
Healthcare Provider	Sign up, sign in, manage assigned patients, monitor communication activity, manage patient phrases and vocabulary, view practice progress, and update professional settings.
Admin	Sign in, manage default global phrases, and manage default global vocabulary.

3.2 General User Flow

Figure UM-1 should show the general Echozy homepage and role selection flow. Users first open the Echozy homepage, click Sign In, choose a role, and continue to the relevant authentication page.

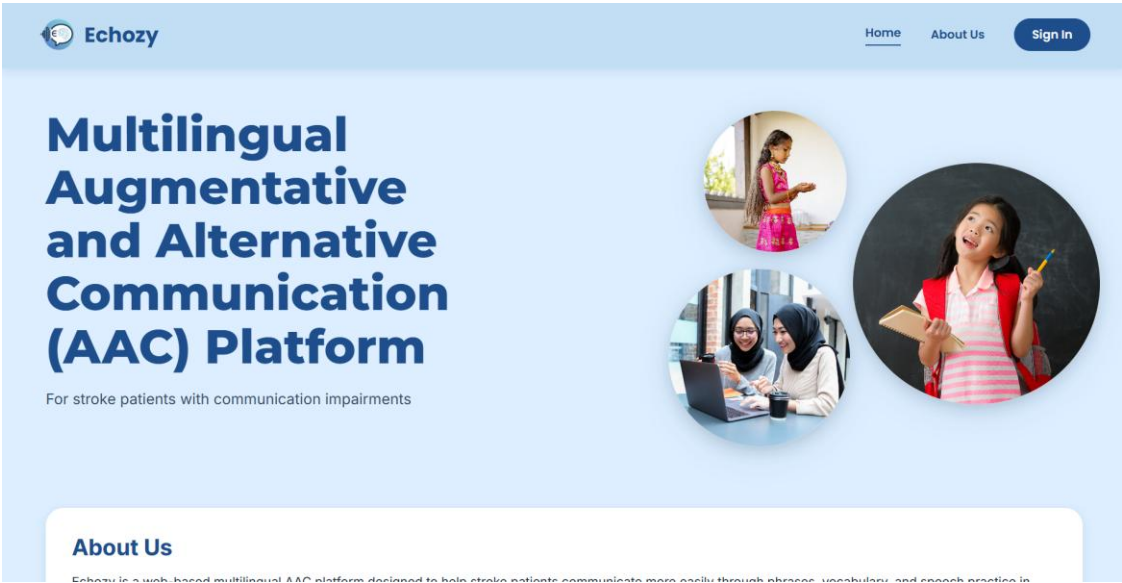


Figure UM-1: Echozy Homepage Flow

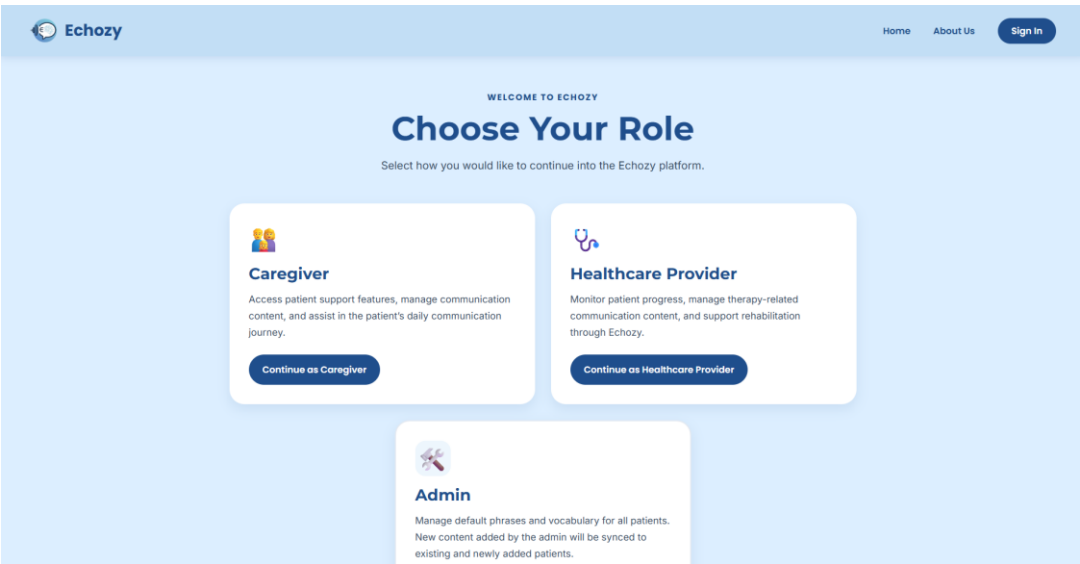


Figure UM-2: Echozy Role Selection Flow

Figure UM-1 and Figure UM-2 shows the starting point of the Echozy user flow, where users access the homepage and select their role before signing in.

1. Open the Echozy website: <https://echozy-aac-platform.vercel.app>
2. Click Sign In on the navigation bar.
3. Choose the correct role: Caregiver, Healthcare Provider, or Admin.
4. Sign up or sign in based on the selected role.
5. After login, the system redirects the user to the correct dashboard.

3.3 Caregiver User Guide

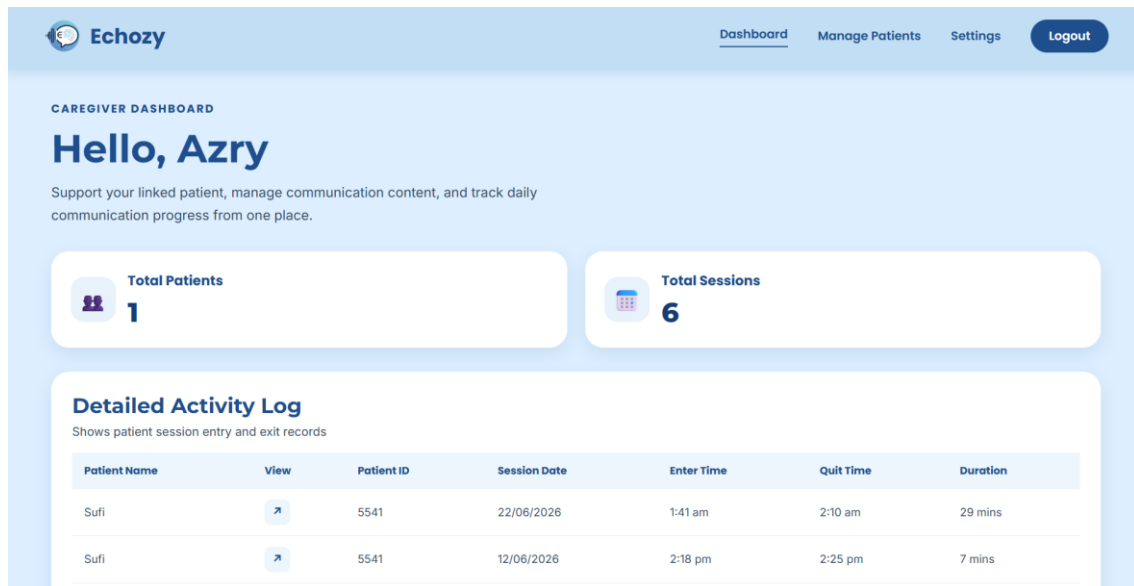


Figure UM-3: Caregiver Dashboard

Figure UM-3 shows the caregiver dashboard. The caregiver dashboard displays summary information such as total patients, total sessions, and detailed activity logs.

1. Open the Echozy website and click Sign In.
2. Choose Caregiver.
3. If the caregiver does not have an account, click Sign Up and enter the required details.
4. Sign in using registered email and password.
5. After successful login, the caregiver dashboard is displayed.
6. Click Manage Patients to add or view patient records.
7. Select a patient card to open the patient dashboard.
8. Manage patient phrases and vocabulary if customization is required.
9. Click Enter Session to start the patient communication session.
10. Click Quit Session after finishing the patient session.

3.4 Healthcare Provider User Guide

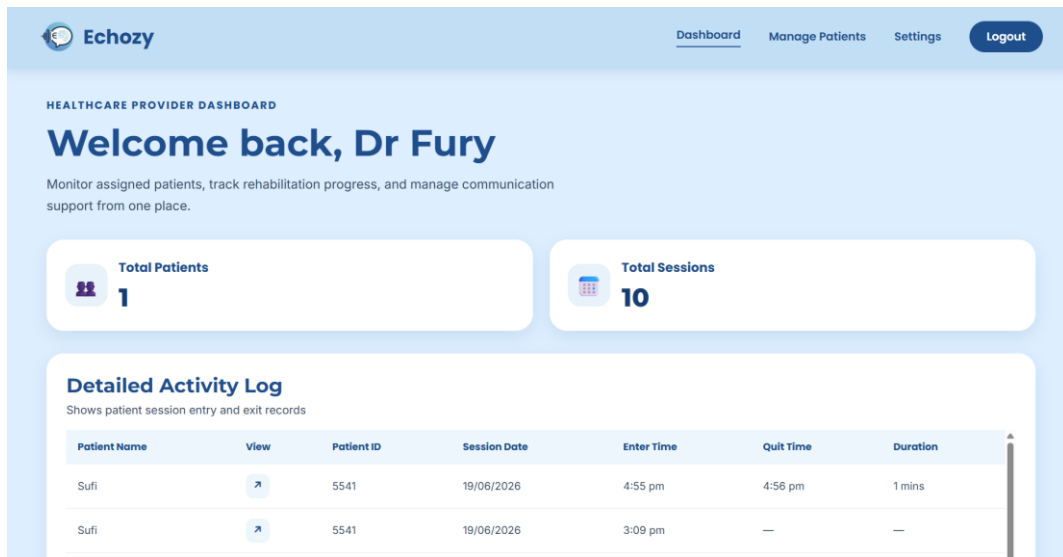


Figure UM-4: Healthcare Provider Dashboard

Figure UM-4 shows the healthcare provider dashboard. Healthcare providers can monitor patient session activity, manage assigned patients, and review patient communication progress.

1. Open the Echozy website and click Sign In.
2. Choose Healthcare Provider.
3. If the healthcare provider does not have an account, click Sign Up and enter full name, staff ID, email, and password.
4. Sign in using registered email and password.
5. Open the dashboard to view total patients, total sessions, and activity logs.
6. Click Manage Patients to add, view, or select patient records.
7. Open a selected patient dashboard to view patient details, frequently used content, and practice progress.
8. Manage patient phrases and vocabulary if needed.
9. Enter and quit patient sessions to record session activity.

3.5 Admin User Guide

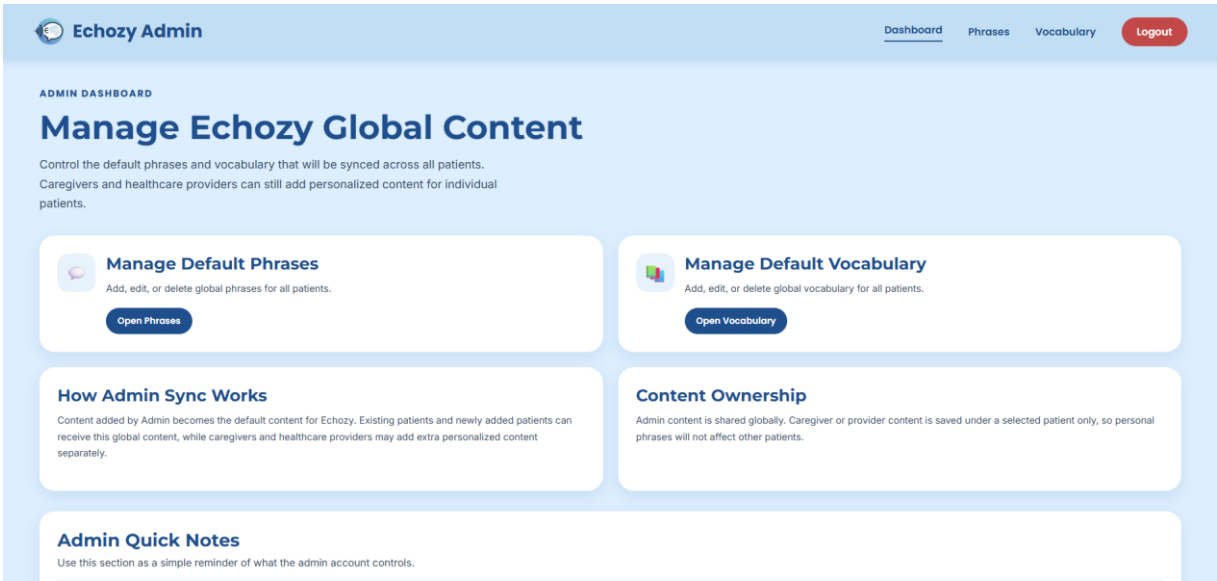


Figure UM-5: Admin Dashboard

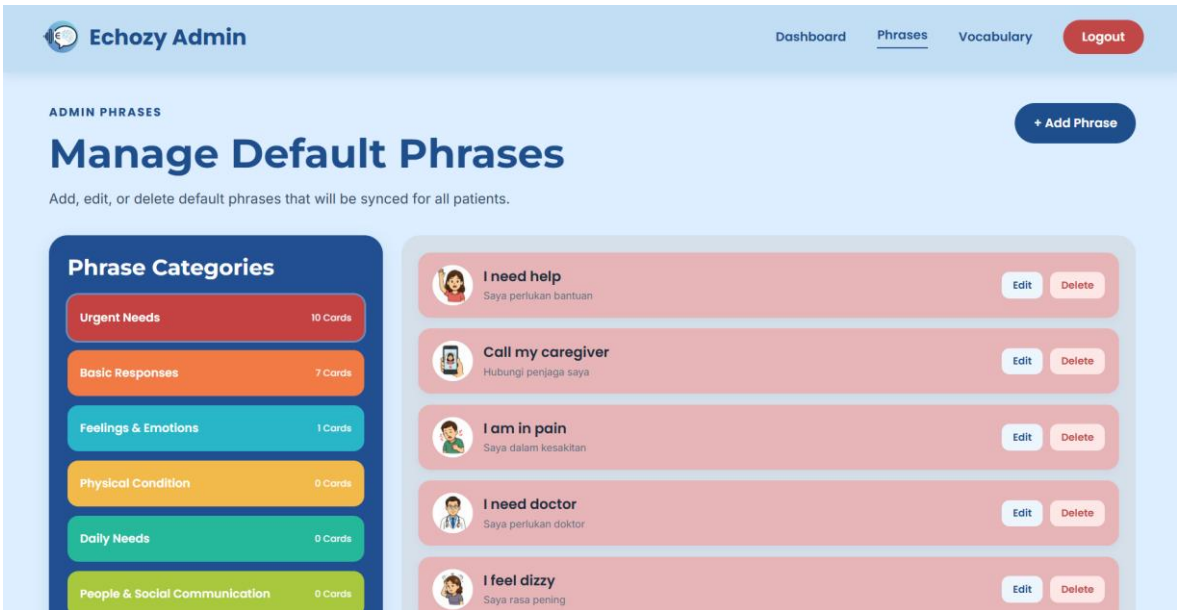


Figure UM-6: Admin Default Phrases

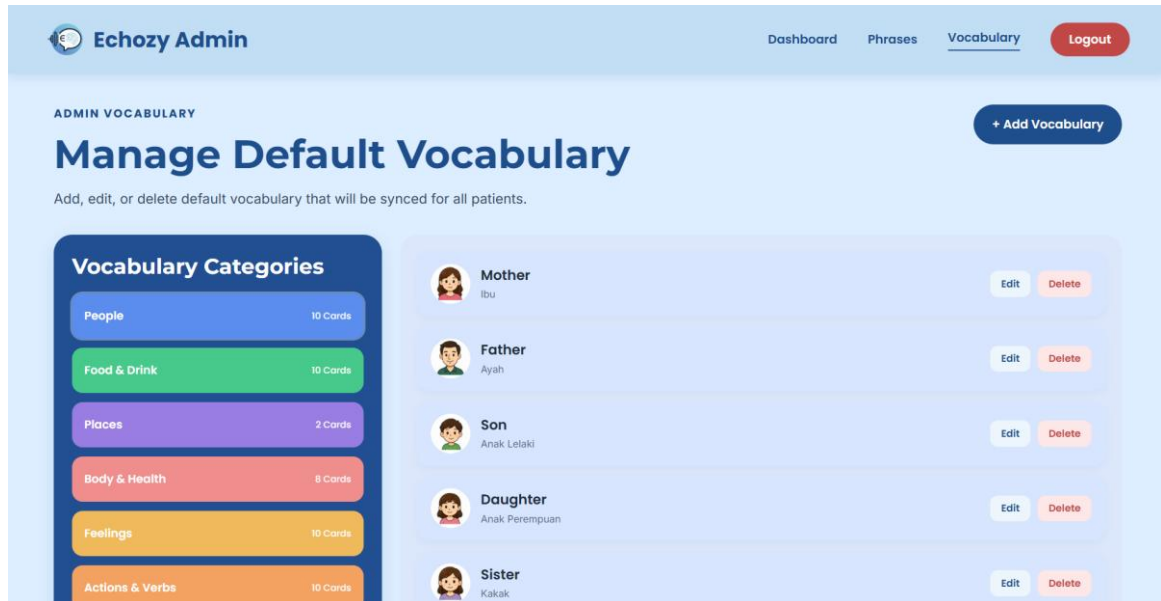


Figure UM-7: Admin Default Vocabulary

Figure UM-5 until Figure UM-7 shows the admin dashboard and default content management pages. Admins manage global phrase and vocabulary content that can be shared across the system.

1. Open the admin sign-in page.
2. Enter the admin email and password.
3. Click Sign In.
4. Open Manage Default Phrases to add, edit, or delete global phrase content.
5. Open Manage Default Vocabulary to add, edit, or delete global vocabulary content.
6. Upload suitable images for phrase or vocabulary cards if needed.
7. Check that the updated content appears correctly in the related phrase or vocabulary list.

3.6 Patient Session User Guide

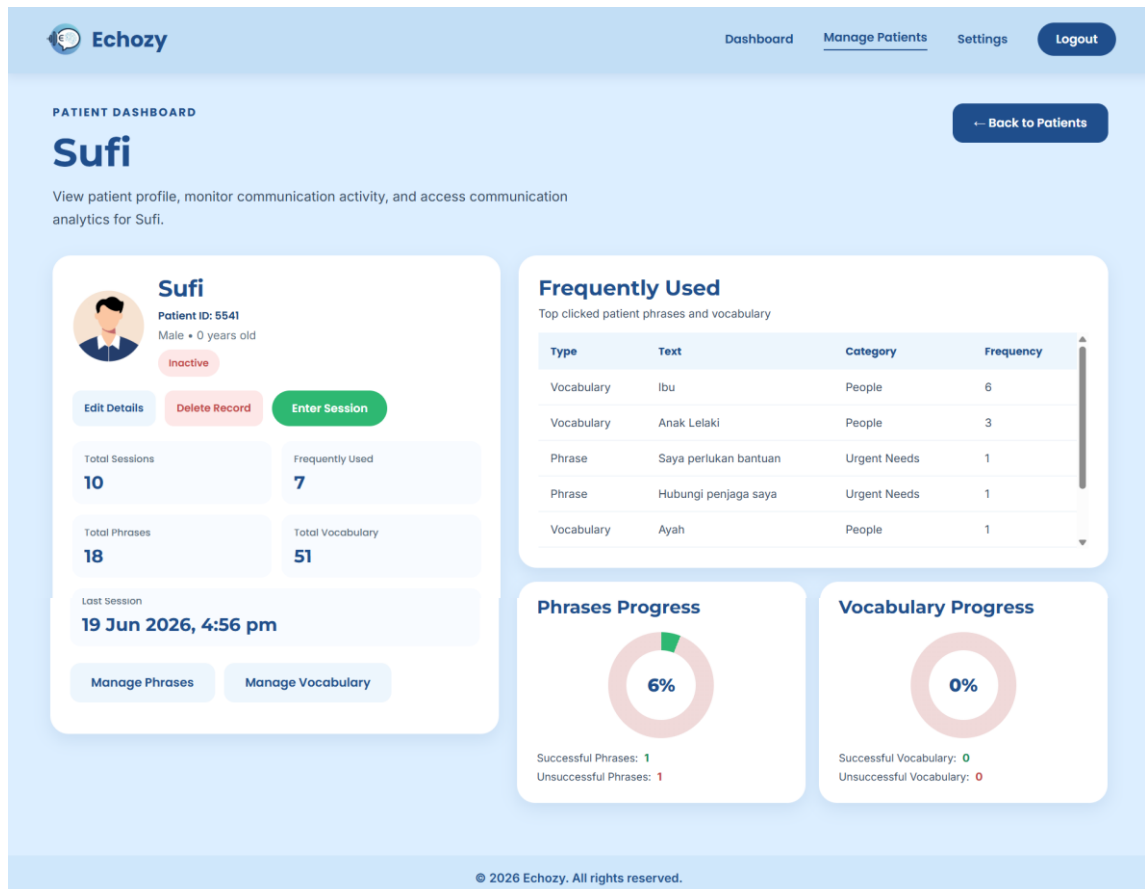


Figure UM-8: Patient Dashboard

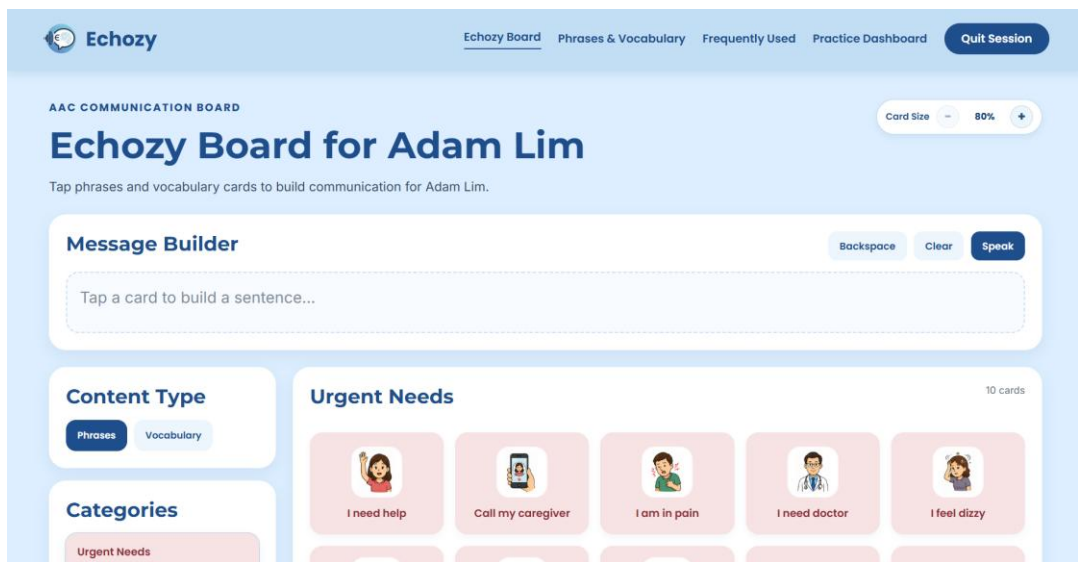


Figure UM-9: Echozy Board

Figure UM-8 and Figure UM-9 shows the patient dashboard and Echozy Board. The patient dashboard provides access to the patient session, while the Echozy Board is used to select communication cards and generate speech output.

1. Caregiver or healthcare provider selects a patient from the Manage Patients page.
2. The system opens the patient dashboard.
3. Click Enter Session to begin the patient session.
4. The patient uses the Echozy Board as the main communication page.
5. Tap phrase or vocabulary cards to build a message.
6. Use Backspace to remove the latest selected item.
7. Use Clear to remove all selected text from the message builder.
8. Click Speak to generate audio output from the selected message.
9. Click Quit Session when the session is complete.

3.7 Frequently Used and Practice Communication

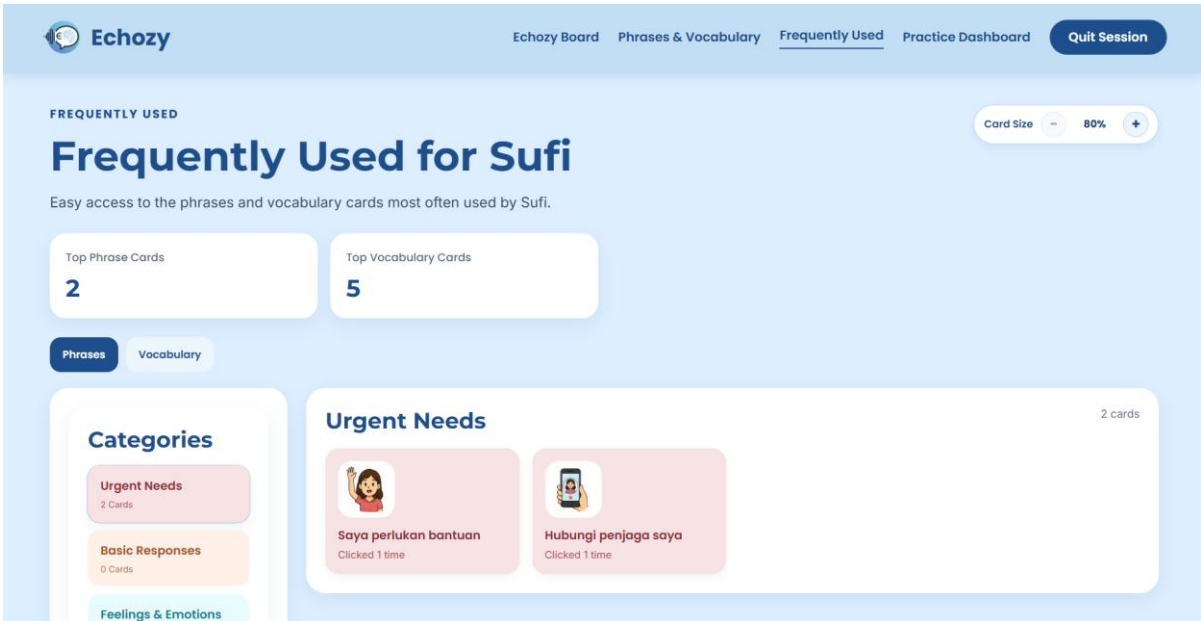


Figure UM-10: Frequently Used Page

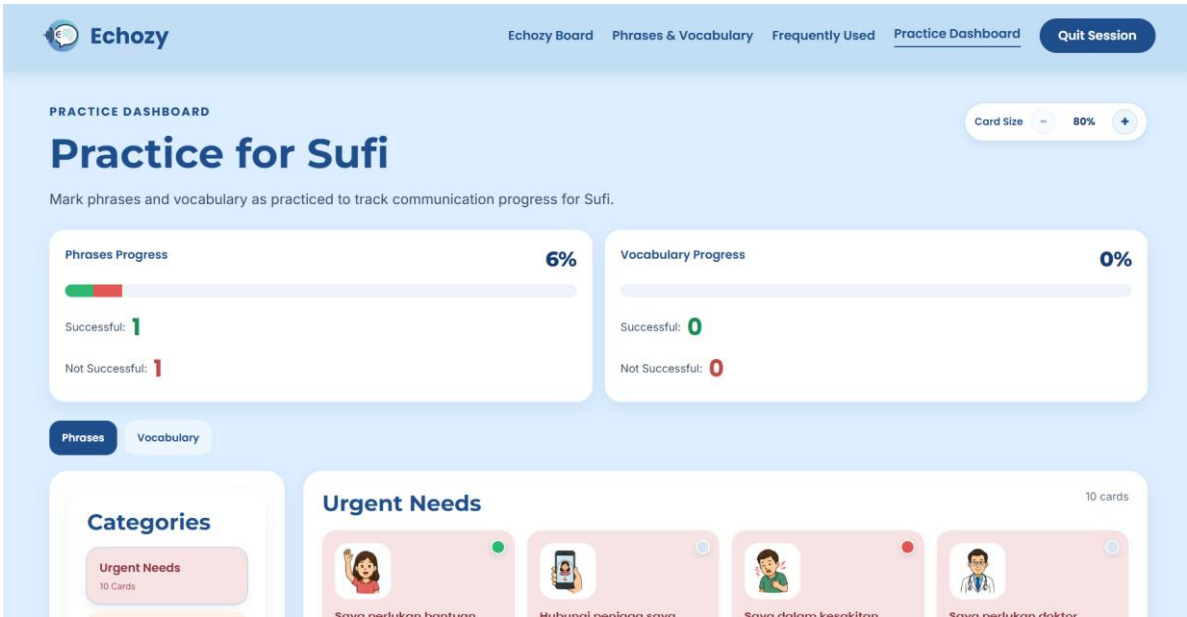


Figure UM-11: Practice Dashboard

Figure UM-10 and Figure UM-11 shows the Frequently Used page and Practice Dashboard. The Frequently Used page displays content selected often by the patient, while the Practice Dashboard supports communication practice progress tracking.

To use the Frequently Used page, open the patient session and click Frequently Used from the side navigation. The system displays commonly selected phrase and vocabulary cards based on usage count records.

To use the Practice Dashboard, open the patient session and click Practice Dashboard. The system displays phrase and vocabulary progress and allows practice results to be recorded as Matched, Almost Match, or No Match, depending on the implemented practice flow.

3.8 Settings and Logout



[Dashboard](#)

[Manage Patients](#)

[Settings](#)

[Logout](#)

ACCOUNT SETTINGS

Caregiver Settings

Manage your caregiver profile, account preferences, and notification settings.



Azry

Caregiver

Email

sufyyy17@gmail.com

Linked Patients

0 Patients

Account Type

Caregiver

Profile Information

Full Name

Azry

Email Address

sufyyy17@gmail.com

Phone Number

0172145614

Gender

Male

Change Password

Current Password

Enter current password

New Password

Enter new password

Confirm New Password

Confirm new password

Reset

Save Changes

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Figure UM-12: Caregiver Settings Page

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Echozy Dashboard Manage Patients Settings Logout

ACCOUNT SETTINGS

Healthcare Provider Settings

Manage your provider profile, organization information, account preferences, and notification settings.

 **Dr Fury**
Healthcare Provider

Email
ffarhanah03@gmail.com

Assigned Patients
0 Patients

Account Type
Healthcare Provider

Department

Profile Information

Full Name
Dr Fury

Email Address
ffarhanah03@gmail.com

Phone Number
Enter your phone number

Gender
Female

Professional Information

Organization / Hospital
UNIMAS

Department
Enter department

Position / Role
Enter position or role

License / Staff ID
HP12345

Change Password

Current Password
Enter current password

New Password
Enter new password

Confirm New Password
Confirm new password

Reset Save Changes

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Figure UM-13: Healthcare Provider Settings Page

Figure UM-12 and Figure UM-13 shows the settings pages for caregivers and healthcare providers. Users can update profile information and change password details from the settings page.

1. Click Settings from the dashboard side navigation.
2. Update the required profile information.
3. For healthcare providers, update professional information such as organization, department, position, or staff ID if needed.
4. Click Save Changes to store the updated information.
5. Click Logout to end the current session safely.

CHAPTER 4

MAINTENANCE NOTES

The following notes are recommended for future maintainers of Echozy:

- ❖ Do not expose the OpenAI API key in frontend JavaScript files.
- ❖ Keep the OpenAI API key inside Supabase secrets or secure backend environment variables.
- ❖ Do not expose Supabase service role keys in frontend code.
- ❖ Use GitHub commits to track every system update.
- ❖ Check Vercel deployment status after pushing updates to GitHub.
- ❖ Monitor OpenAI API usage because online text-to-speech may involve cost over long-term usage.
- ❖ Back up Supabase database records and storage files before making major changes.
- ❖ Update the README.md file when folder structure, setup steps, or technology usage changes.
- ❖ PWA and offline-first support are future improvements and are not part of the current final implementation.

CHAPTER 5

REFERENCES AND USEFUL LINKS

The following links are useful for maintaining, running, or publishing the Echozy system:

Reference	URL	Use
Echozy GitHub Repository	https://github.com/eyfafajay/echozy-aac-platform	Project source code and releases.
Echozy Hosted Website	https://echozy-aac-platform.vercel.app	Published web application.
Supabase Docs	https://supabase.com/docs	Supabase platform documentation.
Supabase Auth Docs	https://supabase.com/docs/guides/auth	Authentication setup and user management.
Supabase Database Docs	https://supabase.com/docs/guides/database/overview	PostgreSQL database setup and management.
Supabase Storage Docs	https://supabase.com/docs/guides/storage	File and image storage setup.
Supabase Edge Functions Docs	https://supabase.com/docs/guides/functions	Backend function setup and deployment.
Supabase Secrets Docs	https://supabase.com/docs/guides/functions/secrets	Secure environment variable storage for Edge Functions.
OpenAI Text-to-Speech Docs	https://platform.openai.com/docs/guides/text-to-speech	Speech generation documentation.
OpenAI API Keys	https://platform.openai.com/api-keys	API key management.
OpenAI API Pricing	https://openai.com/api/pricing/	TTS cost monitoring and planning.
Vercel Git Deployment Docs	https://vercel.com/docs/git	Deployment using GitHub repository.
GitHub Releases Docs	https://docs.github.com/en/repositories/releasing-projects-on-github/about-releases	Release tag and version publishing guide.